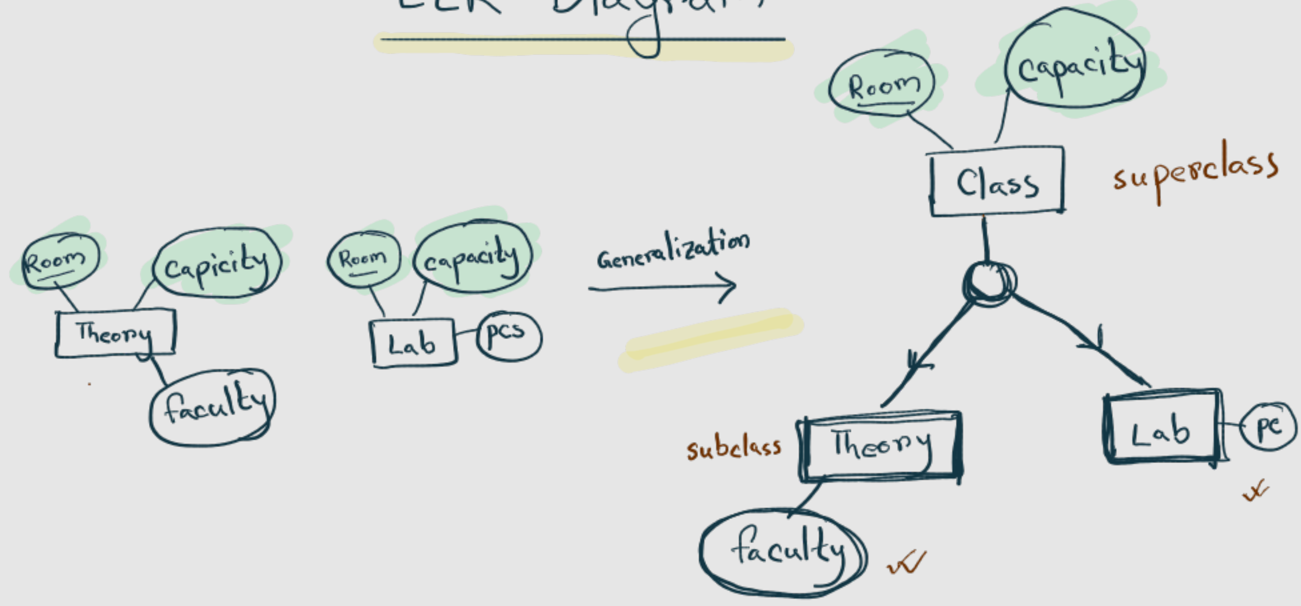
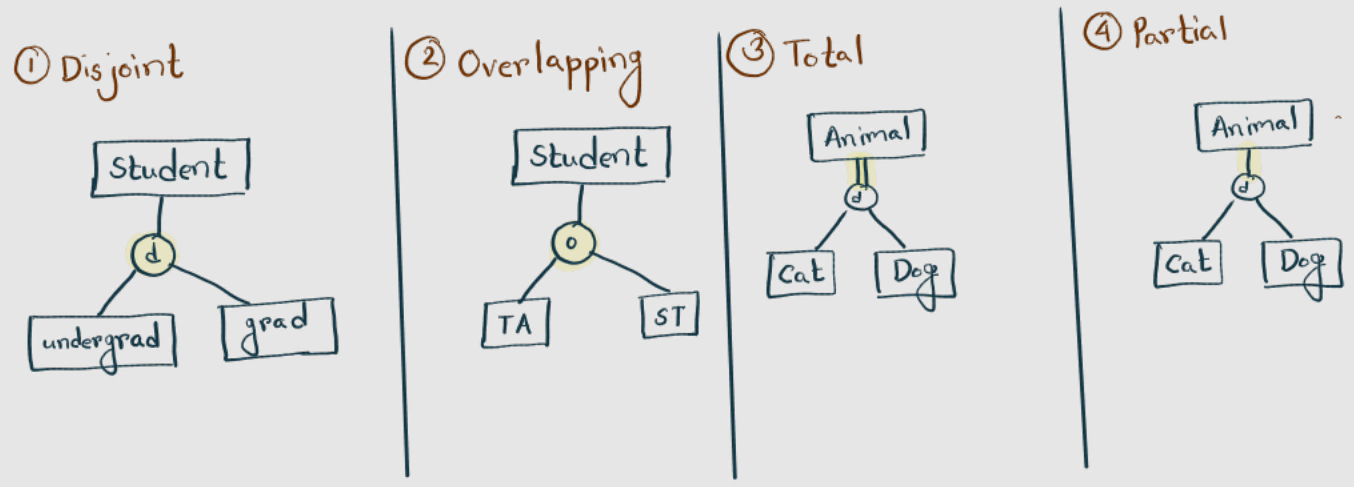
# EER Diagram




## Applying EER Concepts in Practice (1)

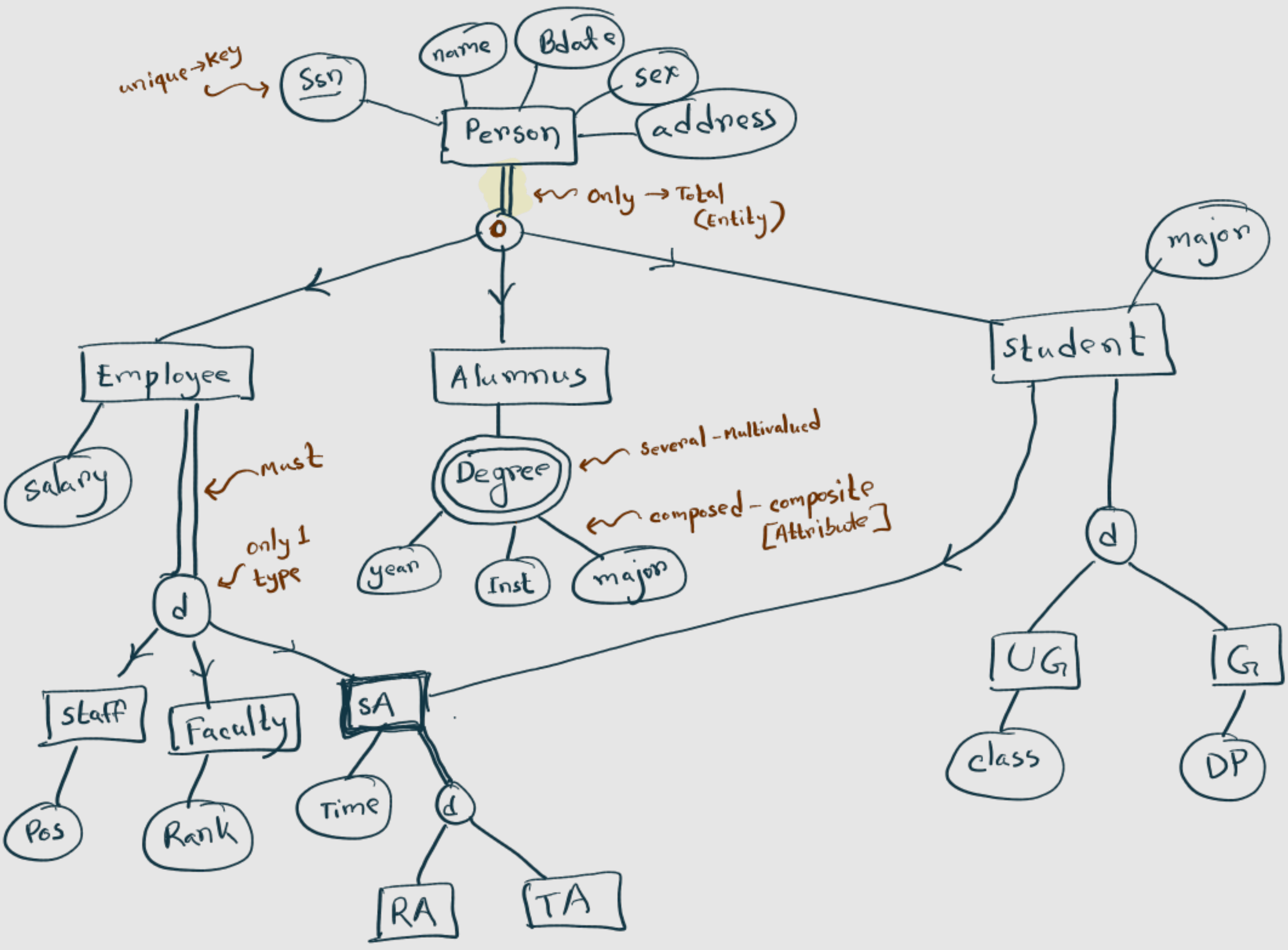
Given the following data requirements for a University database management system, construct an EER diagram:

→ There are only 3 types of persons in the university: Employee, Alumnus and Student. Every person has a unique SSN, Name, Sex, Address, Birth date. Employees have a salary attribute and students have a major attribute. Alumnus has several degrees composed of year, institute and major. A person can be of different types at the same time, e.g. an employee can also be an alumni member.

→ Employees must belong to only 1 type: Staff (having attribute Position), Faculty (having attribute Rank) and Student Assistant (having attribute Percent_Time). Student Assistant also inherits from Students.

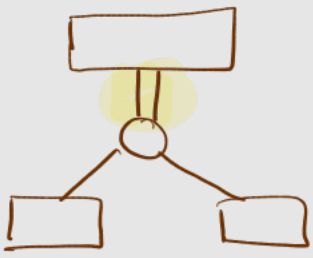
→ Students can be in any one of only two types: undergraduate student or graduate student. Graduate students have Degree_program and undergraduate students have class attributes.

→ Student Assistants can be any one of only two types: Research and Teaching assistant with attributes project and course respectively.

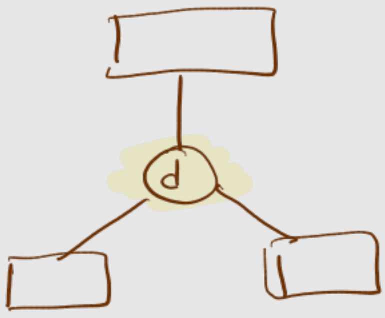


* Unique → key attribute


* Only/Must → Total



* Only 1 type → Disjoint




* Several → Multivalued